GDCh Research Article Angewandte International Edition Chemie www.angewandte.org

How to cite: *Angew. Chem. Int. Ed.* **2025**, *64*, e202513342
doi.org/10.1002/anie.202513342

Asymmetric Catalysis

Enantioselective Synthesis of Cyclobutane-fused Heterocycles via Lewis Acid-Catalyzed Dearomative [2+2] Photocycloaddition of Indoles, Benzofurans, and Benzothiophenes with Alkenes

Muzi Li[+], Andrey D. Kobelev[+], Zhiping Wang[+], Yuan-Jun Gao, Yuan-Zheng Cheng,* Valentine P. Ananikov,* and Shu-Li You**

Abstract: Cyclobutane-fused heterocycles are important motifs in biologically active molecules, yet their enantioselective synthesis remains a significant challenge. We report a broadly applicable and modular strategy for constructing these strained architectures through a visible-light-mediated, Lewis acid-catalyzed dearomative [2+2] photocycloaddition of indoles, benzofurans, and benzothiophenes with alkenes. The method employs a simple catalytic system based on commercially available rare-earth Lewis acids and chiral pyridine-2,6-bis(oxazoline) (PyBox) ligands. A wide array of heteroarenes and styrenes bearing diverse functional groups participate efficiently, delivering cyclobutane-fused products in up to 96% yield, >20:1 dr, and >99% ee. The synthetic utility is further demonstrated by gram-scale synthesis and facile removal of the directing group to access functionalized amino acid derivatives. Mechanistic investigations, including ultraviolet-visible (UV–vis) spectroscopy, nonlinear effect studies, kinetic isotope effects, and density functional theory (DFT) calculations, reveal that a triplet-state heteroarene engages in regio- and enantio-selective C─C bond formation under mild photochemical conditions. This study highlights the potential of excited-state Lewis acid catalysis in unlocking enantioselective dearomatization pathways for complex molecular architectures.

Introduction

Exploring the chemical space of small molecules plays a key role for modern drug discovery program.[1–3] Specifically, three-dimensional (3D) architectures, characterized by a high proportion of sp^3-hybridized carbon atoms, provide ample opportunities to enhance the druggability of candidate molecules through adjusting their absorption, distribution, metabolism, and excretion (ADME) properties.[4] In this context, structural modifications of indolines, one of the most prevalent scaffolds in naturally occurring bioactive alkaloids, typically result in improved biological activity, selectivity, and metabolic stability.[5–8] In light of this, enantioselective synthesis of 3, 5, or 6-membered-ring-fused indolines has garnered increasing attention and achieved significant advancements,[9–12] owing to their favorable affinities for the 3D configurations of various proteins. However, chiral cyclobutane-fused indolines, which possess high ring strain energy, remain largely underexplored in medicinal chemistry due to the absence of efficient synthetic methodologies (Scheme 1a). Moreover, oxygen/sulfur-containing saturated heterocycles also constitute a significant proportion of small molecule drugs.[13–15] Unfortunately, their challenging synthesis has thus far limited their application in pharmaceuticals.

Asymmetric dearomatization of indoles, benzofurans, and benzothiophenes[16–23] has emerged as an indispensable synthetic strategy for building cyclobutane-fused heterocycles. Pioneering work by Bandini and coworkers demonstrated the potential of gold catalysis through an enantioselective [2+2] cycloaddition of indoles with allenamides, which efficiently generated methylenecyclobutane-fused indolines.[24] Sigman, Toste, and their coworkers realized a gold-catalyzed tandem rearrangement/cyclization of indole propargyl acetates by leveraging acyclic diaminocarbene ligands to achieve precise stereocontrol.[25] Notably, photochemical [2+2] cycloaddition reactions have gained prominence due to their inherent atom- and step-economy for assembling these strained architectures.[26–34] However, the unbridled reactivity of the excited-triplet-state organic compounds presented long-standing challenges in stereochemical regulation. This impasse was overcome through Lewis acid-mediated chromophore activation,[35–38] the merger of photocatalysis with asymmetric catalysis [39–45] and other approaches. In particular, the Meggers group employed a Δ-RhS complex to facilitate the visible-light-mediated asymmetric

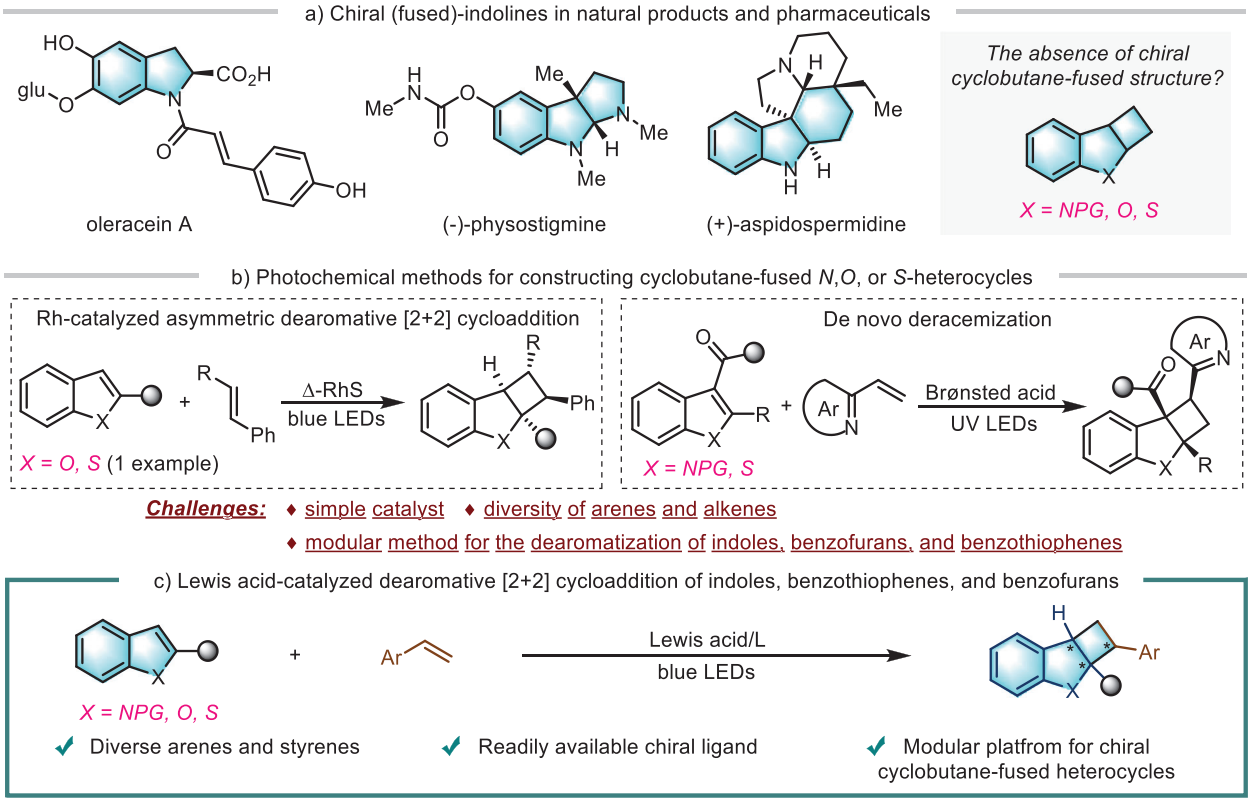
[*] M. Li[+], Z. Wang[+], Prof. Dr. Y.-J. Gao, Prof. Dr. Y.-Z. Cheng, Prof. Dr. S.-L. You
New Cornerstone Science Laboratory, State Key Laboratory of Organometallic Chemistry, Shanghai Institute of Organic Chemistry, University of Chinese Academy of Sciences, Chinese Academy of Sciences, 345 Lingling Lu, Shanghai 200032, China
E-mail: yuanjun.gao@sioc.ac.cn
chengyz@sioc.ac.cn
slyou@sioc.ac.cn

A. D. Kobelev[+], Prof. Dr. V. P. Ananikov
Zelinsky Institute of Organic Chemistry, Russian Academy of Sciences, Leninsky Prospect, 47, Moscow 119991, Russia
E-mail: val@ioc.ac.ru

[+] These authors contributed equally to this work.

Additional supporting information can be found online in the Supporting Information section

© 2025 Wiley-VCH GmbH



Scheme 1. The cyclobutane-fused heterocycles and synthetic methods. a) Valuable indoline derivatives. b) Known asymmetric synthetic methods for cyclobutane-fused heterocycles. c) Modular Lewis acid catalyst-based approach in this work.

[2+2] cycloaddition between benzofurans/benzothiophenes and alkenes (Scheme 1b, left).[46] The Glorius group disclosed a dearomative [2+2] cycloaddition/ring-expansion cascade between indoles and alkenes[47] and the Feng group subsequently achieved enantioselective control of this transformation via a chiral $Tb(OTf)_3$ catalyst.[48] Further advancements include the photoenzymatic intramolecular cycloaddition of indole-alkene by the Chen, Zhong, and Wu groups[49,50] and the Brønsted acid-catalyzed deracemization strategy using 2-vinylazaarenes as diradical acceptors under UV irradiation by the Jiang group (Scheme 1b, right).[51] While these landmark achievements have significantly expanded the synthetic toolbox, critical challenges persist.[52] The development of minimalist catalytic systems allowing asymmetric dearomatization reactions for diverse heteroarenes, remains an urgent priority in synthetic chemistry. Addressing these limitations could unlock novel avenues for drug discovery through streamlined access to novel polycyclic scaffolds.

Building upon our previous discovery of Lewis acid/PyBox-catalyzed asymmetric dearomative [4+2] photocycloaddition of naphthalene derivatives,[53] we sought to probe the untapped potential of this catalytic paradigm in visible-light-mediated dearomative [2+2] cycloaddition. Specifically, we envisioned leveraging the dual capability of Lewis acids to simultaneously coordinate indoles/benzofurans/benzothiophenes and orchestrate stereochemical control, thereby enabling modular synthesis of enantioenriched cyclobutane-fused heterocycles (Scheme 1c). The noteworthy challenges include: 1) managing intermolecular radical addition of excited heteroarenes to alkenes within spatially congested environment; 2) overcoming the heightened ring strain inherent to cyclobutane formation compared to 5 or 6-membered analogues; and 3) achieving convergent control over regio-, diastereo-, and enantio-selectivities in a single catalytic manifold. Herein, we present the detailed investigation of this study.

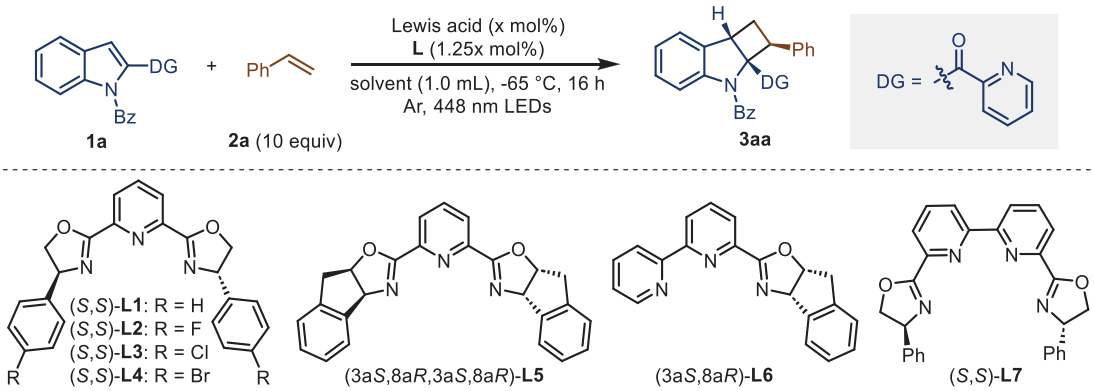
Results and Discussion

Initially, our study was commenced by employing Bz-protecting indole **1a** and styrene **2a** as model substrates to probe the reaction conditions. When **1a** and **2a** were stirred in THF at −65 °C for 16 h in the presence of $Gd(OTf)_3$ and PyBox **L1** under blue LEDs irradiation (λ_{max} = 448 nm), cyclobutane-fused indoline **3aa** was obtained in quantitative yield with 3.3:1 dr and 85%/86% ee (Table 1, Entry 1). To further enhance the diastereo- and enantio-selectivities, a series of ligands were scrutinized, and **L5** emerged as the optimal choice, delivering the desired product in >99% NMR yield with 11.5:1 dr and 97% ee (Entries 2–7). Reducing the loading of the Lewis acid from 20 to 10 mol% had a negligible impact on the reaction outcomes (Entry 8). Other Lewis acids, such as $Tb(OTf)_3$ and $Y(OTf)_3$, were also evaluated but failed to surpass the performance of

© 2025 Wiley-VCH GmbH

15213773, 2025, 42, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/anie.202513342 by Zhejiang University, Wiley Online Library on [01/02/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Table 1: Optimization of the reaction conditions.[a)]



Entry	Lewis acid (x)	L	Solvent	Yield (%)[b)]	dr[b)]	ee (%)[c)]
1	$Gd(OTf)_3$ (20)	**L1**	THF	>99	3.3:1	−85/−86[d)]
2	$Gd(OTf)_3$ (20)	**L2**	THF	>99	2.4:1	−91/−89
3	$Gd(OTf)_3$ (20)	**L3**	THF	>99	1.7:1	−91/−92
4	$Gd(OTf)_3$ (20)	**L4**	THF	>99	2.0:1	−90/−88
5	$Gd(OTf)_3$ (20)	**L5**	THF	>99	11.5:1	97
6	$Gd(OTf)_3$ (20)	**L6**	THF	63	7.1:1	96
7	$Gd(OTf)_3$ (20)	**L7**	THF	94	4.0:1	−69/−90
8	**$Gd(OTf)_3$ (10)**	**L5**	**THF**	**>99**	**12.6:1**	**96**
9	$Tb(OTf)_3$ (10)	**L5**	THF	39	1.8:1	65/−29
10	$Y(OTf)_3$ (10)	**L5**	THF	10	17.3:1	48
11	$Gd(OTf)_3$ (10)	**L5**	EtOAc	>99	12.7:1	93
12	$Gd(OTf)_3$ (10)	**L5**	DCM	88	12.2:1	90
13[e)]	**$Gd(OTf)_3$ (5)**	**L5**	**THF**	**>99(93)**[f)]	**19.5:1**	**98**
14[e),g)]	$Gd(OTf)_3$ (5)	**L5**	THF	>99	14.1:1	97
15[e)]	–	–	THF	24	>20:1	–
16[e)]	$Gd(OTf)_3$ (5)	–	THF	56	1.2:1	–
17[e),h)]	$Gd(OTf)_3$ (5)	**L5**	THF	–	–	–
18[e),i)]	$Gd(OTf)_3$ (5)	**L5**	THF	54	16.4:1	76

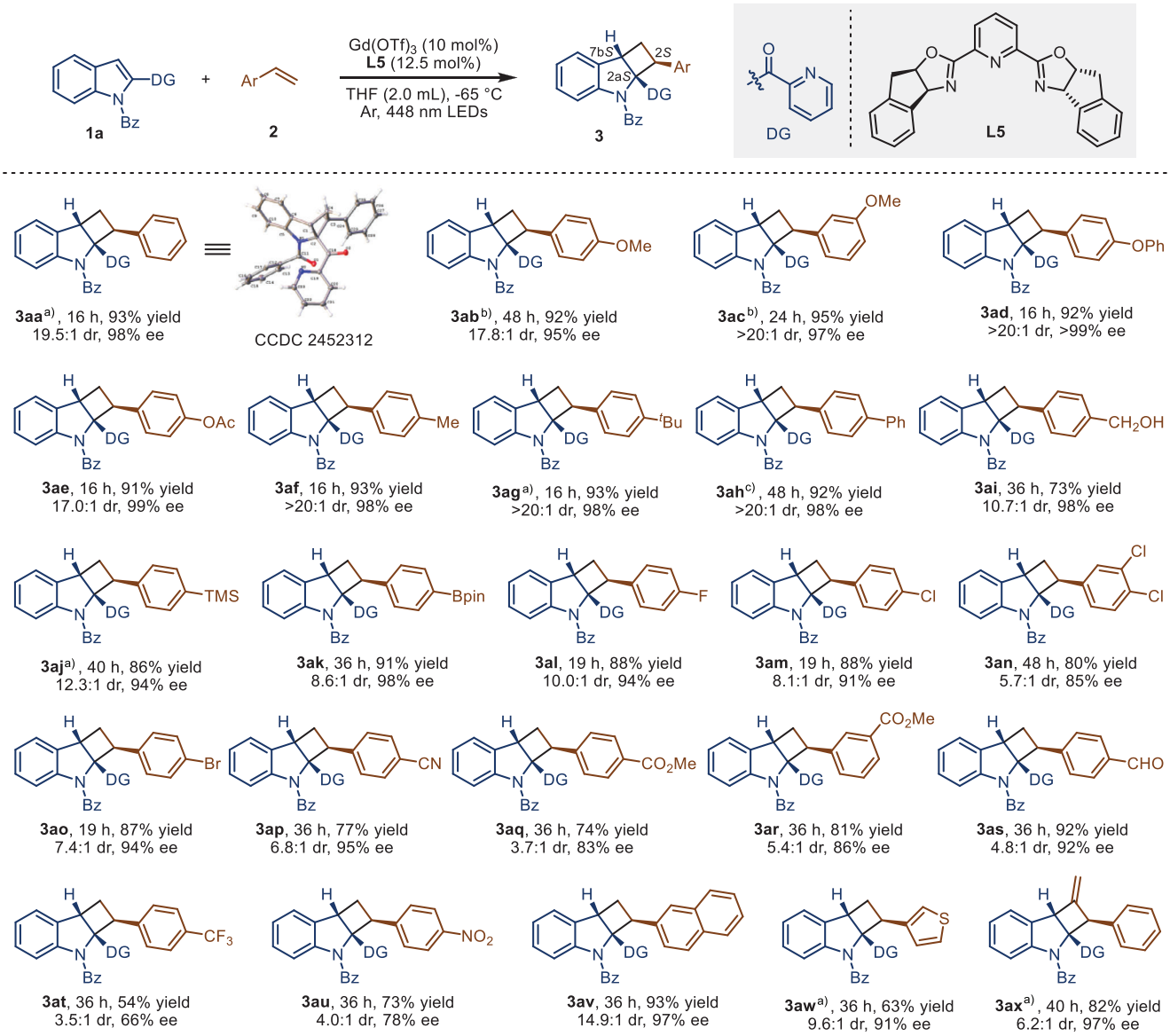
[a)] Reaction conditions: **1a** (0.1 mmol), **2a** (1.0 mmol), Lewis acid (x mol%), **L** (1.25x mol%) in solvent (1.0 mL) at −65 °C under argon atmosphere with blue LEDs. [b)] Determined by ^1H NMR analysis of the crude reaction mixture using CH_2Br_2 as an internal standard. [c)] Determined by HPLC analysis. [d)] A negative ee value indicates the absolute configuration of major enantiomer is reversed. [e)] 0.2 mmol scale, THF (0.4 mL). [f)] Isolated yield. [g)] **2a** (5.0 equiv.). [h)] In dark. [i)] In the presence of air. ee = enantiomeric excess. dr = diastereomeric ratio. THF = tetrahydrofuran. EtOAc = ethyl acetate. DCM = dichloromethane.

$Gd(OTf)_3$ in terms of yield and selectivity (Entries 9 and 10). Subsequently, the solvent investigation revealed that EtOAc and DCM were both suitable, affording **3aa** in excellent results (Entry 11, EtOAc, >99% NMR yield, 12.7:1 dr, 93% ee; Entry 12, DCM, 88% NMR yield, 12.2:1 dr, 90% ee). Increasing the reaction concentration (overall concentration of all dissolved components, from 0.1 to 0.5 M for **1a**) led to slightly improved results, even with a reduced loading of $Gd(OTf)_3$ to 5 mol% (Entry 13). Interestingly, a similar trend was observed when the loading of **2a** was decreased from 10 to 5 equiv. (Entry 14). Control experiments in the absence of the catalyst and light underscored their indispensability for this reaction (Entries 15 and 17). The dramatic decline in the yield of **3aa** to 56% in the absence of the ligand highlighted the significant ligand acceleration effect in this process (Entry 16). The reaction's sensitivity to air supported the involvement of a triplet intermediate (Entry 18). The protecting group Bz in **1a** is crucial for preventing ring expansion following the [2+2] cycloaddition process (for more details, see Table S1). Ultimately, the optimal reaction conditions were established as follows: **1a** (0.1 mmol), **2a** (1.0 mmol), $Gd(OTf)_3$ (10 mol%), **L5** (12.5 mol%), THF (0.1 M), −65 °C, 16 h (Entry 8, >99% NMR yield, 96% ee, 12.6:1 dr), or **1a** (0.2 mmol), **2a** (2.0 mmol), $Gd(OTf)_3$ (5 mol%), **L5** (6.25 mol%), THF (0.5 M), −65 °C, 16 h (Entry 13, >99% NMR yield, 98% ee, 19.5:1 dr).

With the optimal conditions in hand, we began to examine the substrate scope for this enantioselective dearomative [2+2] photocycloaddition reaction. Generally, a broad array of styrenes bearing diverse functional groups were found to be compatible with the reaction conditions (Scheme 2). Styrenes substituted by electron-donating groups such as OMe, OPh, OAc, Me, ^{t}Bu, and Ph on the benzene ring reacted smoothly with **1a**, affording their dearomatized products **3ab**-**3ah** in excellent yields (91%–95%) with excellent diastereoselectivity (17.0:1–>20:1 dr) and enantioselectivity (95%–>99% ee). The CH_2OH, TMS and Bpin groups were well tolerated in this reaction, delivering the desired products **3ai**-**3ak** in 73%–91% yields with 8.6:1–12.3:1 dr and 94%–98% ee. The tolerance of these functional groups offers a valuable opportunity

© 2025 Wiley-VCH GmbH

15213773, 2025, 42, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/anie.202513342 by Zhejiang University, Wiley Online Library on [01/02/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License



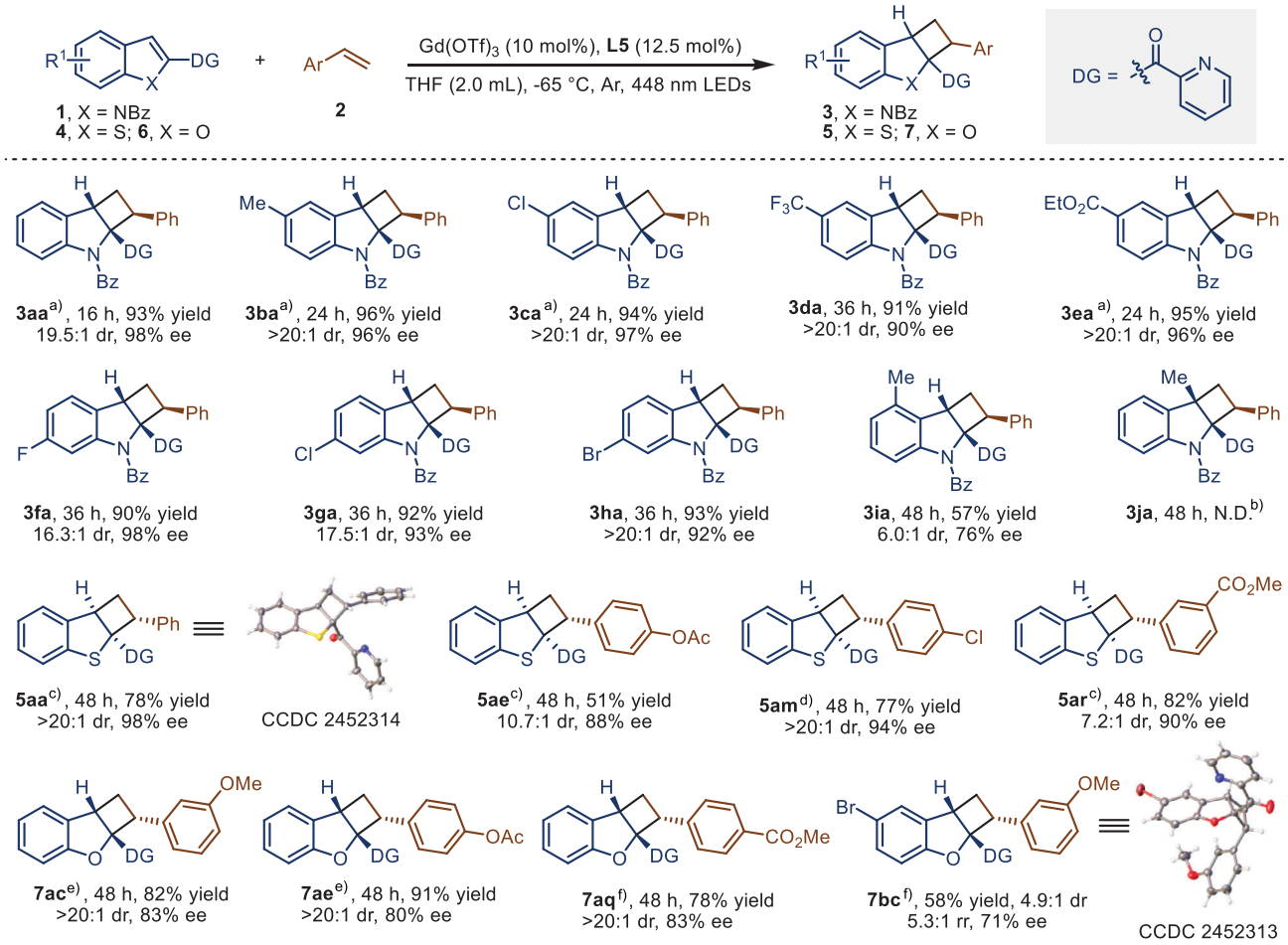
Scheme 2. Scope of Styrenes. Reaction conditions: **1a** (0.2 mmol), **2** (2.0 mmol), $Gd(OTf)_3$ (10 mol%), **L5** (12.5 mol%) in THF (2.0 mL) at −65 °C under argon atmosphere with blue LEDs. Isolated yields were reported. The dr values were determined by ^1H NMR analysis of crude reaction mixtures. The ee values were determined by HPLC analysis. [a] $Gd(OTf)_3$ (5 mol%), **L5** (6.25 mol%), THF (0.4 mL). [b] THF (1.0 mL). [c] **2h** (6.0 equiv.).

for further diversification. This reaction also accommodated styrenes bearing electron-withdrawing groups, such as F, Cl, Br, CN, CO_2Me, and CHO on the benzene ring, affording their desired products **3al**-**3as** in 74%–92% yields with 3.7:1–10.0:1 dr and 83%–95% ee. More electron-deficient CF_3 and NO_2 groups were compatible with the reaction conditions, albeit with inferior results (CF_3, **3at**, 54% yield, 3.5:1 dr, 66% ee; NO_2, **3au**, 73% yield, 4.0:1 dr, 78% ee). Importantly, this reaction was successfully extended to vinylnaphthalene and vinylthiophene, providing the cyclobutane-fused indolines **3av** (93% yield, 14.9:1 dr, 97% ee) and **3aw** (63% yield, 9.6:1 dr, 91% ee), respectively. Additionally, 1-phenylallene was proven to be a feasible substrate, generating methylenecyclobutane **3ax** in 82% yield with 6.2:1 dr and 97% ee. The absolute configuration of **3aa** was established as (2*S*,2a*S*,7b*S*) through X-ray crystallographic analysis of its enantiopure sample. The stereochemistry of other indolines was assigned analogously.

Then, the scope of heteroarenes was evaluated with styrene **2a** (Scheme 3). First, an investigation was conducted on indoles substituted with a diverse array of functional groups. The substituents such as Me, Cl, CF_3, and CO_2Et at the 5 position, as well as F, Cl, and Br at the 6 position, were all well-tolerated in this reaction, affording their desired products **3ba**-**3ha** in 90%–96% yields with 16.3:1–>20:1 dr and 90%–98% ee. The substituents at the 3 or 4 position of indole significantly restrained the reaction efficiency, probably due to their substantial steric hindrance (**3ia**, 4-Me, 57% yield, 6.0:1 dr, 76% ee; **3ja**, 3-Me, N.D.). The reaction of benzothiophene derivative with styrene **2a** by employing **L4** instead of

© 2025 Wiley-VCH GmbH

15213773, 2025, 42, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/anie.202513342 by Zhejiang University, Wiley Online Library on [01/02/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License



Scheme 3. Scope of Heteroarenes. Reaction conditions: **1**, **4** or **6** (0.2 mmol), **2** (2.0 mmol), $Gd(OTf)_3$ (10 mol%), **L5** (12.5 mol%) in THF (2.0 mL) at −65 °C under argon atmosphere with blue LEDs. Isolated yields were reported. The dr values were determined by ^1H NMR analysis of crude reaction mixtures. The ee values were determined by HPLC analysis. [a] $Gd(OTf)_3$ (5 mol%), **L5** (6.25 mol%), THF (0.4 mL). [b] N.D. = not detected. [c] **L4** instead of **L5**. [d] **L3** instead of **L5**. [e] $Tb(OTf)_3$ (10 mol%), **L7** (12.5 mol%), 2-MeTHF (1.0 mL). [f] $Tb(OTf)_3$ (10 mol%), **L7** (12.5 mol%), 2-MeTHF (2.0 mL).

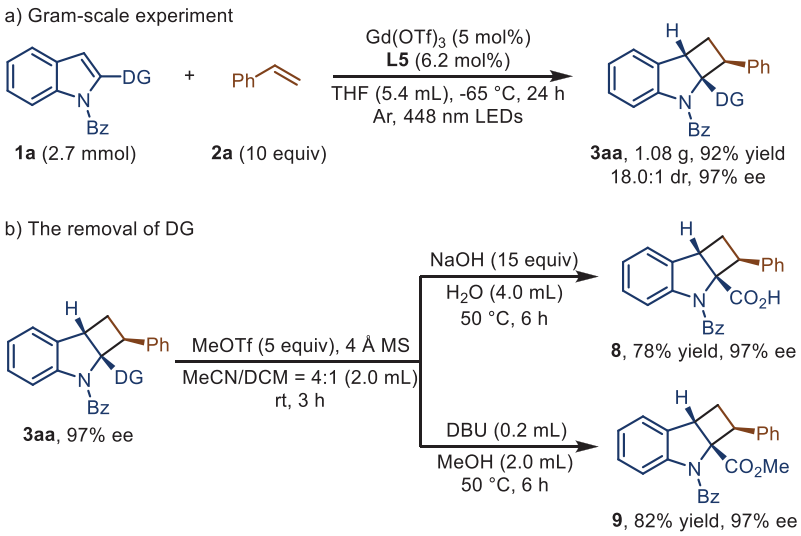
L5 as the ligand, led to the formation of cyclobutane-fused dihydrobenzo[b]thiophene **5aa** in 78% yield with >20:1 dr and 98% ee. The absolute configuration of **5aa** was established as (2*R*,2a*R*,7b*R*) through X-ray crystallographic analysis of its enantiopure sample. Furthermore, styrenes bearing electron-donating group (OAc) and electron-withdrawing groups (Cl, CO_2Me) were found to be viable substrates, yielding the desired products (**5ae**, **5am**, **5ar**) in 51%–82% yields with 7.2:1–>20:1 dr and 88%–94% ee. For the reaction of **5am**, **L3** was employed to enhance the regioselectivity. Moreover, the dearomative [2+2] photocycloaddition reactions of benzofuran derivatives with styrenes were realized using a Lewis acid catalyst composed of $Tb(OTf)_3$ and **L7**. This approach afforded cyclobutane-fused dihydrobenzofurans **7ac**, **7ae**, **7aq**, and **7bc** in 58%–91% yields with 4.9:1–>20:1 dr and 71%–83% ee. The absolute configuration of **7bc** was established as (2*R*,2a*S*,7b*S*) through X-ray crystallographic analysis of its enantiopure sample. The successful application of this Lewis acid catalytic system to benzothiophenes and benzofurans highlights its broad utility as a unified platform for visible-light-driven enantioselective dearomatization of heteroarenes.

To demonstrate the synthetic utility of this dearomatization protocol, a gram-scale experiment and the removal of the directing group were conducted (Scheme 4). The reaction of **1a** (2.7 mmol) with **2a** under the optimal conditions gave **3aa** in 92% yield with 18.0:1 dr and 97% ee, which were comparable to those of 0.2 mmol-scale reaction (Scheme 4a). Subsequently, the removal of acylic pyridine from **3aa** was achieved through a methylation/substitution cascade procedure in the presence of H_2O or MeOH (Scheme 4b).[54,55] This process readily furnished the conformationally rigid amino acid **8** and its methyl ester **9** in 78% and 82% yields, respectively, without the loss of the stereochemical integrity. The resulting carboxylic acid moiety constitutes a privileged and readily diversifiable handle that effectively alleviates the substrate-scope constraints imposed by an obligatory directing group.

Next, several mechanistic studies were performed to shed lights on the reaction mechanism (Figure 1). Initially, when (*Z*)-*β*-deuterium styrene **2a**-d_1 was reacted with **1a** under the standard conditions, *trans*-**3aa** was obtained as the predominant product (*trans*:*cis* = 1.8:1), demonstrating the involvement of excited triplet indole in this reaction

© 2025 Wiley-VCH GmbH

15213773, 2025, 42, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/anie.202513342 by Zhejiang University, Wiley Online Library on [01/02/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License



Scheme 4. a) Gram-scale experiment. b) Removal of directing group.

(Figure 1a).[56] Furthermore, the *Z*/*E* ratio (13.8:1) of the recovered **2a**-d_1 indicated that the radical addition of triplet indole to styrene was reversible. In the UV–vis absorption spectroscopy, the addition of Gd(OTf)$_3$/**L5** to a THF solution of **1a** resulted in a conspicuous bathochromic shift (Figure 1b). Notably, a more pronounced red shift was observed when only Gd(OTf)$_3$ was added to the solution of **1a**, initially suggesting the potential for a robust racemic background reaction. However, the control experiment in the absence of **L5** underscored its pivotal role in accelerating this reaction (Table 1, entry 16). It was hypothesized that the coordination mode between **1a** and Lewis acid would be modulated by the introduction of ligand, thereby altering the photophysical properties of the Gd(III) complex.[47] The ee of **3aa** was found to be linearly correlated with the ee of **L5**, implying that likely one ligand molecule is incorporated into each active Gd(III) complex (Figure 1c).[57] To elucidate the rate-determining step, secondary kinetic isotope effect (KIE) experiments were conducted.[58] However, the competitive KIE value of 1.05 and parallel KIE value of 1.04 were significantly larger than the typical inverse KIE (<1) (Figure 1d). In addition, the $^{12}C/^{13}C$ KIEs at natural abundance were determined by quantitative ^{13}C NMR experiments for the reaction of **1a** with **2a** (two parallel runs at 13.1% and 11.6% conversions).[59] Strong KIEs were observed for both carbon atoms of the new C3─C4 bond formed via the radical addition step (~1.041 for C3 and ~1.042 for C4), whereas no significant KIEs were detected for the C2─C5 bond formed via radical combination (Figure 1e). These findings suggest that the radical addition of triplet indole to styrene is likely the rate-determining step.

To understand the reaction mechanism of Gd(III)-catalyzed dearomative [2+2] photocycloaddition of indoles, we carried out DFT calculation in Gaussian16 software and plotted the energy profile in Figure 2. D3 dispersion correction scheme of Grimme was used to descript the empirical van der Waals correction at the M06-2X/6-311+G**//B3LYP/6-31G* level. The reaction between **1a** and **2a** by Gd(III) derived catalyst bearing three OTf anions and one ligand **L5** was employed as the computational model. The complex of the Gd(III)-catalyst with **1a** was set as the zero-point of the potential energy surface (**INT1**, 0.0 kcal mol^{-1}). In **INT1**, both nitrogen and oxygen atoms from the directing group of **1a** were coordinated to the Gd(III) center, forming an octet stable structure. The excitation energy of **INT1** from the ground state to the first triplet state (**INT1-T$_1$**) was calculated to be 46.9 kcal mol^{-1}, which meant that **INT1** could be excited to the T$_1$ state under the irradiation of the blue light (448 nm). The spin densities of C1, C2, and C3 atoms were 0.583, 0.281, and 0.337 in **INT1-T$_1$**, implying that the unpaired electrons mainly focused on these positions and the C1 atom had a priority to attack the terminus of styrene.

The first C─C bond-forming step exhibited the highest energy barrier throughout the entire reaction pathway and was identified as the rate-determining step, consistent with the experimental KIE data. During this process, styrene approached the *Re* or *Si* face of indole moiety while adopting two different orientations. Therefore, there were four transition states: **TS1-*Re1*** (53.2 kcal mol^{-1}), **TS1-*Re2*** (51.6 kcal mol^{-1}), **TS1-*Si1*** (49.7 kcal mol^{-1}), and **TS1-*Si2*** (49.1 kcal mol^{-1}). It indicated that the Gibbs free energies of **TS1-*Re*** were 1.9 kcal mol^{-1} higher than those of **TS1-*Si***. The geometric variance between **TS1-*Re*** and **TS1-*Si*** mainly originated from the benzoyl group of indole moiety and **L5** of Gd(III) catalyst. In **TS1-*Re1***, strong steric repulsion between **2a** and the benzoyl group of **1a** weakened the interaction between the styrene and the indole ring. In **TS1-*Re2***, similar steric repulsive effect occurred between **2a** and **L5** group of Gd(III) catalyst. Such unfavorable steric repulsion was not found in **TS1-*Si***. These results confirmed that the installation of a protecting group on the indole nitrogen atom and the utilization of **L5** ligand in the Gd(III) catalyst synergistically were beneficial for high enantioselectivity. Moreover, the energy difference between the competing transition states accounts for the excellent enantioselectivity observed experimentally. The free energy of **TS1-*Si1*** for the main product **3aa** was calculated to be only 0.6 kcal mol^{-1}

© 2025 Wiley-VCH GmbH

15213773, 2025, 42, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/anie.202513342 by Zhejiang University, Wiley Online Library on [01/02/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

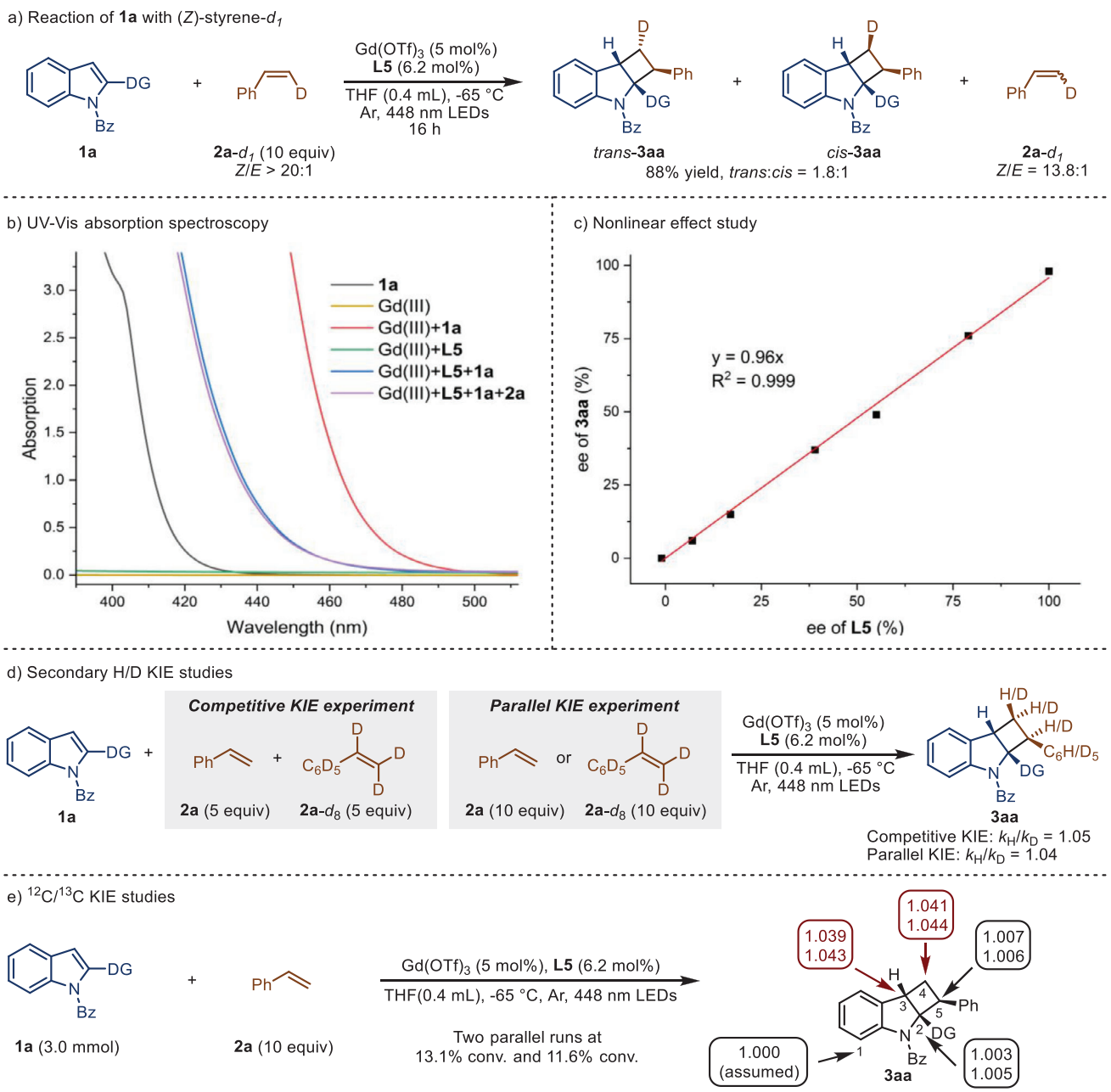


Figure 1. Mechanistic Investigation.

higher than that of **TS1-*Si2***, suggesting that both pathways were accessible but the latter should be preferred. It also supported the deuteration experiment that *trans*-**3aa** was the predominant product (*trans*:*cis* = 1.8:1). Thermodynamically, the intermediate **INT2-*Si1*** (26.6 kcal mol^{-1}) via **TS1-*Si1*** was more stable than **INT2-*Si2*** (27.6 kcal mol^{-1}) via **TS1-*Si2***. The **INT2-*Si2*** was transformed to **INT2-*Si1*** through a rapid C─C σ-bond rotation (**TS2**, 30.1 kcal mol^{-1}). Next, the triplet state **INT2-*Si1*** was converted to its open-shelled singlet state (**INT2-OSS**, 27.8 kcal mol^{-1}) via the minimal energy crossing points **MECP**. Subsequently, the second C–C bond formation facilitated the construction of the cyclobutane intermediate **INT3** (6.2 kcal mol^{-1}). Finally, with the dissociation of desired product **3aa** and the coordination of **1a**, **INT1** would be regenerated, which closed the catalytic cycle. The whole process was endoergic by 0.3 kcal mol^{-1}. Thus, the theoretical analysis provides a mechanistic foundation for the stereochemical control achieved in the reaction and supports the proposed role of the Lewis acid complex in guiding the selectivity of the photocycloaddition process.

Conclusion

In this work, we have developed a unified and modular strategy for the enantioselective synthesis of cyclobutane-fused heterocycles via visible-light-driven dearomative [2+2] photocycloaddition of indoles, benzofurans, and benzothiophenes

© 2025 Wiley-VCH GmbH

15213773, 2025, 42, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/anie.202513342 by Zhejiang University, Wiley Online Library on [01/02/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

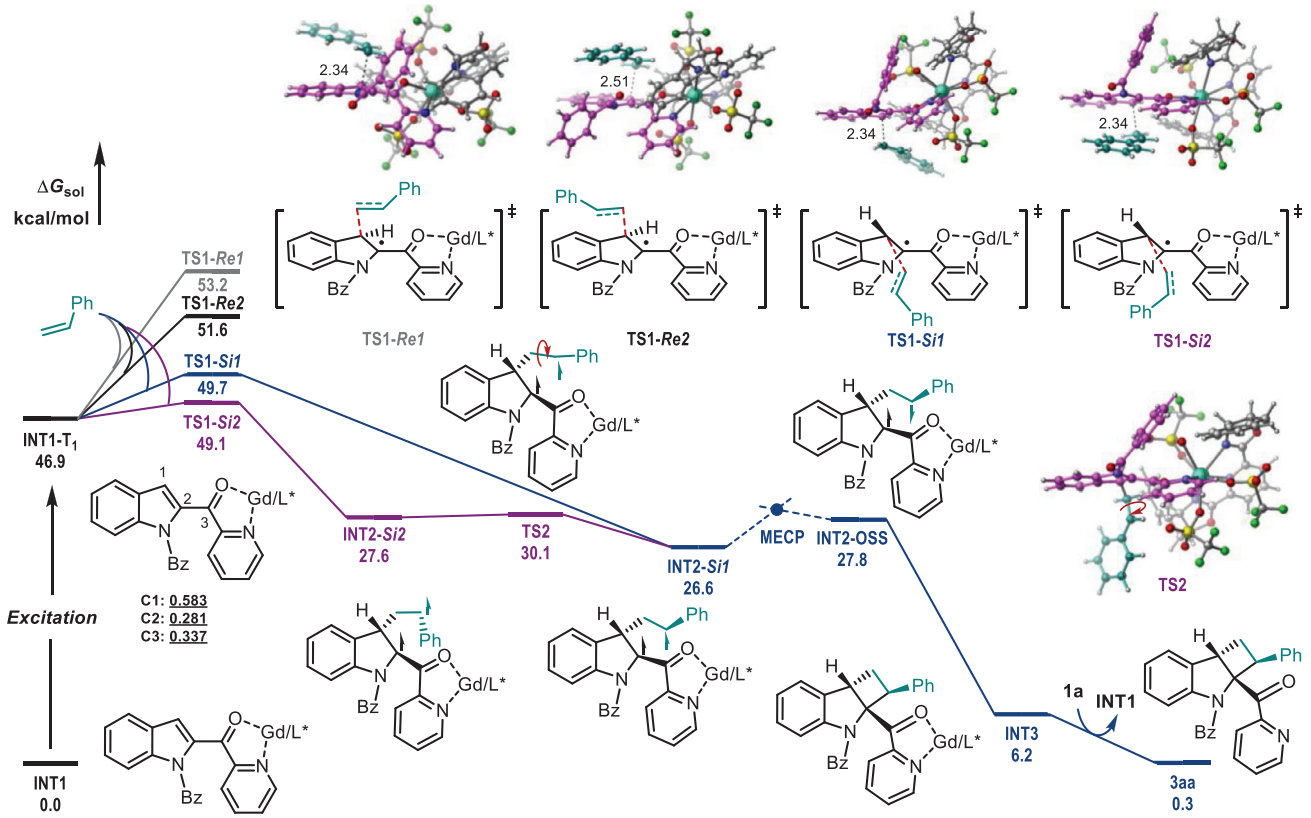


Figure 2. Energy profile for the reaction of **1a** with **2a** by the Gd(III)-catalyst calculated at the M06-2X-D3/6-311+G**//B3LYP-D3/6-31G* level. The Gibbs free energies (ΔG_{sol}) are in kcal mol^{-1} and bond distances are in Å.

with alkenes. This approach employs a readily accessible Lewis acid/chiral PyBox ligand catalytic system that operates under mild conditions with broad substrate scope and excellent stereocontrol. The ability to construct complex, highly strained polycyclic frameworks bearing multiple contiguous stereocenters, particularly enantioenriched cyclobutane-fused indolines, dihydrobenzofurans, and dihydrobenzothiophenes, represents a significant advance in asymmetric dearomatization chemistry. Importantly, this methodology provides access to privileged molecular architectures that are otherwise difficult to obtain, offering a valuable tool for expanding the 3D chemical space relevant to drug discovery. The synthetic utility of the method is demonstrated by gram-scale reactions and efficient downstream functionalization, including the preparation of rigid amino acid derivatives. Mechanistic studies, supported by UV–vis spectroscopy, nonlinear effects, isotope labeling, and DFT calculations, reveal a mechanistically distinct pathway involving triplet-state heteroarenes and stereocontrolled radical additions, offering new insights into the behavior of excited-state Lewis acid catalysis. These results contribute fundamentally to our understanding of asymmetric photochemistry and open a new avenue for controlling reactivity and selectivity in photoinduced dearomatization. Overall, this work not only introduces a powerful synthetic method with direct relevance to medicinal chemistry, but also provides a conceptual framework for future exploration of photoexcited-state transformations under asymmetric catalysis. Further exploration of excited-state Lewis acid catalysis in asymmetric dearomatization reactions is currently underway in our laboratory.

Supporting Information

The authors have cited additional references within the Supporting Information.[60–76]

Acknowledgements

Financial supports for this work were provided by National Key R&D Program of China (2021YFA1500100), National Natural Science Foundation of China (22201291, 22261132511, 22031012, and 22401291), Strategic Priority Research Program of Chinese Academy of Sciences (XDB0610000), Youth Innovation Promotion Association of the Chinese Academy of Sciences (2023000046), Science and Technology Commission of Shanghai Municipality (22JC1401103, 2023000285, 24QA2711600 and 24ZR1478700), and Russian Science Foundation (23-43-00086). S.-L.Y. acknowledges the support from New Cornerstone Science Foundation.

Conflict of Interests

The authors declare no conflict of interest.

© 2025 Wiley-VCH GmbH

15213773, 2025, 42, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/anie.202513342 by Zhejiang University, Wiley Online Library on [01/02/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Data Availability Statement

The data that support the findings of this study are available in the Supporting Information of this article.

Keywords: Asymmetric catalysis · Cyclobutane · Dearomatization · Heterocycle · Photocycloaddition

[1] S. L. Schreiber, *Science* **2000**, *287*, 1964–1969.
[2] W. R. J. D. Galloway, A. Isidro-Llobet, D. R. Spring, *Nat. Commun.* **2010**, *1*, 80.
[3] J.-L. Reymond, R. van Deursen, L. C. Blum, L. Ruddigkeit, *MedChemComm* **2010**, *1*, 30–38.
[4] F. Lovering, J. Bikker, C. Humblet, *J. Med. Chem.* **2009**, *52*, 6752–6756.
[5] M. A. Mohamed, R. S. Mohamed, T. A.-Q. Nadia, K. M. Huda, A. F. Thoraya, *Mini-Rev. Med. Chem.* **2018**, *18*, 1409–1421.
[6] A. Thakur, A. Singh, N. Kaur, R. Ojha, K. Nepali, *Bioorg. Chem.* **2020**, *94*, 103436.
[7] J. N. Marcelo, K. L. Hannah, *Curr. Med. Chem.* **2021**, *28*, 4828–4844.
[8] G.-J. Mei, W. L. Koay, C. X. A. Tan, Y. Lu, *Chem. Soc. Rev.* **2021**, *50*, 5985–6012.
[9] D. Zhang, H. Song, Y. Qin, *Acc. Chem. Res.* **2011**, *44*, 447–457.
[10] W. Zi, Z. Zuo, D. Ma, *Acc. Chem. Res.* **2015**, *48*, 702–711.
[11] F.-T. Sheng, J.-Y. Wang, W. Tan, Y.-C. Zhang, F. Shi, *Org. Chem. Front.* **2020**, *7*, 3967–3998.
[12] Y.-Z. Liu, H. Song, C. Zheng, S.-L. You, *Nat. Synth.* **2022**, *1*, 203–216.
[13] M. D. Delost, D. T. Smith, B. J. Anderson, J. T. Njardarson, *J. Med. Chem.* **2018**, *61*, 10996–11020.
[14] C.-S. Jiang, W. E. G. Müller, H. C. Schröder, Y.-W. Guo, *Chem. Rev.* **2012**, *112*, 2179–2207.
[15] S. Maji, B. Debnath, S. Panda, T. Manna, A. Maity, R. Dayaramani, R. Nath, S. A. Khan, M. J. Akhtar, *Chem. Biodivers.* **2024**, *21*, e202400473.
[16] C.-X. Zhuo, W. Zhang, S.-L. You, *Angew. Chem. Int. Ed.* **2012**, *51*, 12662–12686.
[17] J.-B. Chen, Y.-X. Jia, *Org. Biomol. Chem.* **2017**, *15*, 3550–3567.
[18] C. Zheng, S.-L. You, *Nat. Prod. Rep.* **2019**, *36*, 1589–1605.
[19] A. Cerveri, M. Bandini, *Chin. J. Chem.* **2020**, *38*, 287–294.
[20] G. Zhen, K. Jiang, B. Yin, *ChemCatChem* **2022**, *14*, e202200099.
[21] M. Aleksiev, O. García Mancheño, *Chem. Commun.* **2023**, *59*, 3360–3372.
[22] Y.-M. He, Y.-Z. Cheng, Y. Duan, Y.-D. Zhang, Q.-H. Fan, S.-L. You, S. Luo, S.-F. Zhu, X.-F. Fu, Q.-L. Zhou, *CCS Chem.* **2023**, *5*, 2685–2716.
[23] Y.-Z. Cheng, M. Li, R.-X. Wang, L.-H. Zhu, W.-J. Shen, X.-X. Zou, Q. Gu, S.-L. You, *Prog. Chem.* **2024**, *36*, 1785–1829.
[24] M. Jia, M. Monari, Q.-Q. Yang, M. Bandini, *Chem. Commun.* **2015**, *51*, 2320–2323.
[25] Z. L. Niemeyer, S. Pindi, D. A. Khrakovsky, C. N. Kuzniewski, C. M. Hong, L. A. Joyce, M. S. Sigman, F. D. Toste, *J. Am. Chem. Soc.* **2017**, *139*, 12943–12946.
[26] P. J. Wagner, *Acc. Chem. Res.* **2001**, *34*, 1–8.
[27] J. Iriondo-Alberdi, M. F. Greaney, *Eur. J. Org. Chem.* **2007**, *2007*, 4801–4815.
[28] D. M. Schultz, T. P. Yoon, *Science* **2014**, *343*, 1239176.
[29] S. Poplata, A. Tröster, Y.-Q. Zou, T. Bach, *Chem. Rev.* **2016**, *116*, 9748–9815.
[30] M. Zhu, C. Zheng, X. Zhang, S.-L. You, *J. Am. Chem. Soc.* **2019**, *141*, 2636–2644.
[31] D. Sarkar, N. Bera, S. Ghosh, *Eur. J. Org. Chem.* **2020**, *2020*, 1310–1326.
[32] P. Franceschi, S. Cuadros, G. Goti, L. Dell'Amico, *Angew. Chem. Int. Ed.* **2023**, *62*, e202217210.
[33] E. R. Wearing, Y.-C. Yeh, G. G. Terrones, S. G. Parikh, I. Kevlishvili, H. J. Kulik, C. S. Schindler, *Science* **2024**, *384*, 1468–1476.
[34] Q.-B. Zhang, F. Li, B. Pan, L. Yu, X.-G. Yue, *Chem. - Eur. J.* **2024**, *30*, e202401501.
[35] R. Brimioulle, T. Bach, *Science* **2013**, *342*, 840–843.
[36] X. Huang, T. R. Quinn, K. Harms, R. D. Webster, L. Zhang, O. Wiest, E. Meggers, *J. Am. Chem. Soc.* **2017**, *139*, 9120–9123.
[37] C. Brenninger, J. D. Jolliffe, T. Bach, *Angew. Chem. Int. Ed.* **2018**, *57*, 14338–14349.
[38] H. Yu, S. Dong, Q. Yao, L. Chen, D. Zhang, X. Liu, X. Feng, *Chem. - Eur. J.* **2018**, *24*, 19361–19367, and references therein.
[39] J. Du, K. L. Skubi, D. M. Schultz, T. P. Yoon, *Science* **2014**, *344*, 392–396.
[40] S. Das, C. Zhu, D. Demirbas, E. Bill, C. K. De, B. List, *Science* **2023**, *379*, 494–499.
[41] T. R. Blum, Z. D. Miller, D. M. Bates, I. A. Guzei, T. P. Yoon, *Science* **2016**, *354*, 1391–1395.
[42] Q.-Q. Zhou, Y.-Q. Zou, L.-Q. Lu, W.-J. Xiao, *Angew. Chem. Int. Ed.* **2019**, *58*, 1586–1604.
[43] X. Li, J. Großkopf, C. Jandl, T. Bach, *Angew. Chem. Int. Ed.* **2021**, *60*, 2684–2688.
[44] Z. C. Girvin, L. F. Cotter, H. Yoon, S. J. Chapman, J. M. Mayer, T. P. Yoon, S. J. Miller, *J. Am. Chem. Soc.* **2022**, *144*, 20109–20117.
[45] J. B. Kidd, T. A. Fiala, W. B. Swords, Y. Park, K. A. Meyer, K. M. Sanders, I. A. Guzei, J. C. Wright, T. P. Yoon, *J. Am. Chem. Soc.* **2024**, *146*, 15293–15300.
[46] N. Hu, H. Jung, Y. Zheng, J. Lee, L. Zhang, Z. Ullah, X. Xie, K. Harms, M.-H. Baik, E. Meggers, *Angew. Chem. Int. Ed.* **2018**, *57*, 6242–6246.
[47] J. Ma, F. Schäfers, C. Daniliuc, K. Bergander, C. A. Strassert, F. Glorius, *Angew. Chem. Int. Ed.* **2020**, *59*, 9639–9645.
[48] L. Hou, L. Yang, G. Yang, Z. Luo, W. Xiao, L. Yang, F. Wang, L.-Z. Gong, X. Liu, W. Cao, X. Feng, *J. Am. Chem. Soc.* **2024**, *146*, 23457–23466.
[49] N. Sun, J. Huang, J. Qian, T.-P. Zhou, J. Guo, L. Tang, W. Zhang, Y. Deng, W. Zhao, G. Wu, R.-Z. Liao, X. Chen, F. Zhong, Y. Wu, *Nature* **2022**, *611*, 715–720.
[50] J. Guo, J. Qian, D. Cai, J. Huang, X. Yang, N. Sun, J. Zhang, T. Pang, W. Zhao, G. Wu, X. Chen, F. Zhong, Y. Wu, *J. Am. Chem. Soc.* **2024**, *146*, 19030–19041.
[51] J. Wang, Q. Fu, S. Cao, X. Lv, Y. Yin, X. Ban, X. Zhao, Z. Jiang, *J. Am. Chem. Soc.* **2024**, *146*, 22840–22849.
[52] K. Narasaka, Y. Hayashi, H. Shimadzu, S. Niihata, *J. Am. Chem. Soc.* **1992**, *114*, 8869–8885.
[53] M. Li, X.-L. Huang, Z.-Y. Zhang, Z. Wang, Z. Wu, H. Yang, W.-J. Shen, Y.-Z. Cheng, S.-L. You, *J. Am. Chem. Soc.* **2024**, *146*, 16982–16989.
[54] M. P. Sibi, Y.-H. Yang, *Synlett* **2008**, *2008*, 83–88.
[55] H. Ma, L. Xie, Z. Zhang, L.-g. Wu, B. Fu, Z. Qin, *J. Org. Chem.* **2017**, *82*, 7353–7362.
[56] R. Guo, S. Adak, P. Bellotti, X. Gao, W. W. Smith, S. N. Le, J. Ma, K. N. Houk, F. Glorius, S. Chen, M. K. Brown, *J. Am. Chem. Soc.* **2022**, *144*, 17680–17691.
[57] C. Girard, H. B. Kagan, *Angew. Chem. Int. Ed.* **1998**, *37*, 2922–2959.
[58] M. Gómez-Gallego, M. A. Sierra, *Chem. Rev.* **2011**, *111*, 4857–4963.
[59] D. A. Singleton, A. A. Thomas, *J. Am. Chem. Soc.* **1995**, *117*, 9357–9358.
[60] N. V. Hanhan, N. R. Ball-Jones, N. T. Tran, A. K. Franz, *Angew. Chem. Int. Ed.* **2012**, *51*, 989–992.

© 2025 Wiley-VCH GmbH

15213773, 2025, 42, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/anie.202513342 by Zhejiang University, Wiley Online Library on [01/02/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

[61] K. Demkiw, H. Araki, E. L. Elliott, C. L. Franklin, Y. Fukuzumi, F. Hicks, K. Hosoi, T. Hukui, Y. Ishimaru, E. O'Brien, Y. Omori, M. Mineno, H. Mizufune, N. Sawada, Y. Sawai, L. Zhu, *J. Org. Chem.* **2016**, *81*, 3447–3456.
[62] C.-X. Liu, P.-P. Xie, F. Zhao, Q. Wang, Z. Feng, H. Wang, C. Zheng, S.-L. You, *J. Am. Chem. Soc.* **2023**, *145*, 4765–4773.
[63] G. Kirby, L. Grimaud, M. R. Vitale, G. Prestat, F. Berhal, *Green Chem.* **2021**, *23*, 9428–9432.
[64] M. J. Frisch, G. W. Trucks, H. B. Schlegel, G. E. Scuseria, M. A. Robb, J. R. Cheeseman, G. Scalmani, V. Barone, G. A. Petersson, H. Nakatsuji, X. Li, M. Caricato, A. V. Marenich, J. Bloino, B. G. Janesko, R. Gomperts, B. Mennucci, H. P. Hratchian, J. V. Ortiz, A. F. Izmaylov, J. L. Sonnenberg, D. Williams-Young, F. Ding, F. Lipparini, F. Egidi, J. Goings, B. Peng, A. Petrone, T. Henderson, D. Ranasinghe, et al., Gaussian 16, Revision A.03, Gaussian, Inc., Wallingford CT **2016**.
[65] A. D. Becke, *J. Chem. Phys.* **1993**, *98*, 5648–5652.
[66] C. Lee, W. Yang, R. G. Parr, *Phys. Rev. B: Condens. Matter Mater. Phys.* **1988**, *37*, 785–789.
[67] P. J. Stephens, F. J. Devlin, C. F. Chabalowski, M. J. Frisch, *J. Phys. Chem.* **1994**, *98*, 11623–11627.
[68] S. Grimme, J. Antony, S. Ehrlich, H. Krieg, *J. Chem. Phys.* **2010**, *132*, 154104.
[69] D. Andrae, U. Häussermann, M. Dolg, H. Stoll, H. Preuss, *Theor. Chim. Acta* **1990**, *77*, 123–141.
[70] L. E. Roy, P. J. Hay, R. L. Martin, *J. Chem. Theory Comput.* **2008**, *4*, 1029–1031.
[71] F. Weigend, R. Ahlrichs, *Phys. Chem. Chem. Phys.* **2005**, *7*, 3297–3305.
[72] F. Weigend, *Phys. Chem. Chem. Phys.* **2006**, *8*, 1057–1065.
[73] A. V. Marenich, C. J. Cramer, D. G. Truhlar, *J. Phys. Chem. B* **2009**, *113*, 6378–6396.
[74] J. N. Harvey, M. Aschi, H. Schwarz, W. Koch, *Theor. Chem. Acc.* **1998**, *99*, 95–99.
[75] T. Lu, sobMECP program, http://sobereva.com/286 (accessed Sep. 28, **2023**).
[76] C. Y. Legault, CYLview20, Université de Sherbrooke, **2020**, (http://www.cylview.org).

Manuscript received: June 18, 2025
Revised manuscript received: August 17, 2025
Manuscript accepted: August 17, 2025
Version of record online: August 27, 2025

© 2025 Wiley-VCH GmbH

15213773, 2025, 42, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/anie.202513342 by Zhejiang University, Wiley Online Library on [01/02/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License